
Weaponizing the Electromagnetic Spectrum: The PRC's High-powered Microwave Warfare Ambitions

Executive Summary:

The People's Liberation Army (PLA) is rapidly expanding its arsenal of high-power microwave (HPM) weapons as part of its broader strategy to achieve dominance in the electromagnetic spectrum. Recent breakthroughs—including the deployment of mobile-platform HPM systems—signal the PLA's intent to integrate these capabilities into its asymmetric warfare toolkit, enabling disruption of adversary electronic systems.

HPM development in the People's Republic of China (PRC) is closely linked to its evolving doctrine of “cyber-electromagnetic space” warfare. The PLA's emphasis on informatized warfare highlights HPM weapons as a bridge between kinetic and non-kinetic operations, targeting adversaries' command, control, and communication infrastructure.

Strategic lessons from the Russia-Ukraine war and the PLA's own military modernization agenda suggest that HPM capabilities could play a decisive role in future conflicts, including a Taiwan contingency. The PLA is likely to synchronize HPM strikes with cyberattacks to paralyze critical infrastructure, enabling rapid battlefield advantage. This trajectory poses new challenges for the U.S. and its regional allies seeking to protect their C4ISR networks against electronic disruption.

The PRC broke new ground in its high-powered microwave (HPM) technology this past year. At the Zhuhai Air Show in November 2024—a biennial expo that is a major platform for showcasing advances in the aerospace industry—the PLA showcased at least three novel HPM ground-based weapons. These reportedly included the newly designed anti-drone Hurricane 2000 and Hurricane 3000 model HPMs ([Foreign Military Studies Office](#), February 26). The third HPM system was the FK-4000, an anti-drone platform capable of precision single-point attacks and intercepting drone swarms, according to its designer, the China Aerospace Science and Technology Corporation (CASC) ([CASC](#), November 12, 2024). These new weapon systems build upon rapid improvements in HPM industrial-programming software and the development of mobile-platform HPM systems.

The growing investment in HPM technology underscores the People Liberation Army's (PLA) strategic emphasis on asymmetric capabilities. As an asymmetric weapon capable of quickly disabling an adversary's electronic systems, HPMs target the central nodes that hold technologically advanced militaries together, making them a critical weapon in future conflicts. The PLA's pursuit of HPM capabilities aligns with its broader strategic objectives and contributes to its vision of informatized and intelligentized warfare.

PRC Path to Dominating HPM Research and Development

The PRC's interest in electromagnetic pulse (EMP) weapons can be traced back to the Cold War era. U.S. and Soviet nuclear tests in the 1960s revealed that high-altitude nuclear detonations produced EMPs that disabled or destroyed electronics. [1] Following these tests, the PLA strategists were quick to integrate these nuclear EMP weapons into their broader approach to information and cyber warfare ([Science of Military Strategy](#), January 2020). They viewed these weapons as an effective asymmetric capability designed to even the playing field against technologically superior adversaries, such as the United States, which relied on an array of electronics for their increasingly integrated networks of weapons and communication equipment.

Recent PLA research has shifted toward non-nuclear EMP technologies, particularly high-power microwave (HPM) weapons, to achieve the same battlefield effects without nuclear escalation. HPMs are directed energy weapons that emit concentrated high-frequency EMPs at a target to damage and disable its electronic systems. In 2017, HPM development made headlines in the People's Republic of China (PRC) as the State Council awarded Huang Wenhua (黄文华) the highest national award for scientific development for his work on HPMs. Huang, the deputy director of the PRC's Northwest Institute of Nuclear Technology, led a team that reportedly designed an HPM weapon capable of being armed as a ship-borne antimissile weapon ([U.S.-China Economic and Security Review Commission](#), November 14, 2019).

The PRC now dominates HPM research globally. A RAND report estimated that, as of 2022, 90 percent of all new HPM-related patents globally belong to PRC-affiliated researchers and organizations ([RAND](#), April 24, 2024). The level of development in this field is notable, exemplified by the release of Yaoguang, a newly open-sourced HPM development software, created by the University of Electronic Science and Technology of China. According to a study published by the university, Yaoguang can perform full 3D radiation pattern simulations in just 12 minutes, significantly outperforming comparable U.S. tools in computational efficiency. [2] These advancements align with Beijing's emphasis on electronic warfare as part of its broader military modernization agenda. [3]

A Range of Active and Potential HPM Weapons

The new Hurricane 2000, Hurricane 3000, and FK-4000 HPM system showcased at the recent Zhuhai Air Show are deployed on the backs of trucks. The Hurricane models are capable of rapidly rotating and engaging targets in the air above them and reportedly can disable hundreds of drones in short succession with an estimated effective range of 2–3 kilometers ([Foreign Military Studies Office](#),

February 26). Details regarding the FK-4000 HPM's capabilities remain limited. These new weapon systems build upon the HPM device developed by the PRC's Southeast University in 2023, a honeycomb-shaped device capable of emitting multiple EMP beams at distinct targets. [4] Alongside earlier developments, such as Huang Wenhua's 2017 shipborne HPM weapon, these advancements indicate the PLA's expanding strategy to deploy advanced HPMs on mobile platforms.

The PLA maintains a large arsenal of cruise and hypersonic missiles that could potentially deploy HPMs based on similar U.S. weapon systems. These are ideal for first strikes due to their difficult detection and high survivability. The current size of the PLA's cruise missile arsenal is unknown; that said, a 2010 U.S. Department of Defense report estimated the PLA had 200–500 operational HN-2 cruise missiles (DOD, January 1, 2010). In 2019, the PLA unveiled its new hypersonic glide missile system, the medium-range DF-17 (CSIS, April 23, 2024). This missile, traveling at high speeds, can maneuver as it approaches a target, further complicating anti-defense measures.

The PLA could potentially attach their HPMs to its HN-2s or DF-17s, similar to the U.S. military's Counter-electronics High Power Microwave Advanced Missile Project (CHAMP)—an HPM device attached to an AGM-86 cruise missile that can attack seven different targets in under one hour (Air Command and Staff College, 2020). This requires advanced technology to ensure precision strikes from a fast-moving platform. The PRC's advancements in synchronization technologies and their development of mobile-platformed HPMs indicate their capability to develop such weapon systems. The PRC is also heavily investing in EMP shielding technologies to protect its military and civilian infrastructure (Qin Feng et al., September 1, 2023). Some studies focus on PLA operational platforms, including research on EMP shielding for warships, aircraft, and strategic facilities, ensuring the PLA's resilience in high-intensity electronic warfare environments, emphasizing the importance of maintaining dominance in this domain (Li Yuebo et al., 2023).

HPMs are Critical to PLA Informatized Warfare

The PLA has sought to integrate HPM weapons into its operational doctrine, emphasizing their role in disabling enemy command, control, communications, computers, intelligence, surveillance, and reconnaissance (C4ISR) systems, in addition to serving as an anti-access weapon in aerial and naval spaces (RAND, February 17, 2011). Senior PLA officials and defense strategists have consistently underscored the importance of electromagnetic spectrum dominance (Center for Strategic and Budgetary Assessments, October 5, 2017). The PLA's official publications, such as the 2015 defense white paper "China's Military Strategy" (中国的军事战略), highlight electromagnetic superiority as essential for future conflicts. The military's mouthpiece, PLA Daily, has also explored the strategic evolution of electromagnetic warfare, emphasizing how EMP weapons have become critical to modern warfare as forces become more technologically advanced (Xinhua, May 26, 2015; PLA Daily, May 26, 2022).

HPM weapons are critical for the PLA's strategy of informatized warfare that seeks to dominate the information space through advanced electronic and cyber capabilities. This strategy emphasizes targeting technologically advanced adversaries' deep reliance on electronics, using EMPS to disable their integrated network of C4ISR infrastructure. Based on this strategic outlook, the Central Military Commission (CMC) has issued directives emphasizing the need to enhance the PLA's electromagnetic capabilities, with CMC Chairman Xi Jinping reinforcing the importance of informatized warfare (National Defense Science & Technology, 2022).

Central to the PLA's informatized warfare strategy is its evolving concept of "cyber electromagnetic space" warfare (网络电磁空间作战). The *Science of Military Strategy*, one of the PLA's most authoritative doctrinal publications, dedicates an entire chapter to this integrated domain, emphasizing the fusion of electronic and cyber technologies to conduct multi-layered information attacks (Science of Military Strategy, January 2020). This integrated concept differs from that of the United States, which does not combine the two into a singular domain (Institute for National Defense and Security Research, May 2022). The Institute for National Defense and Security Research states that the PLA's doctrine further diverges from the U.S. conception of cyber warfare in that it incorporates kinetic cyber attacks into its framework to a greater degree, alongside soft attacks. HPMs are seen as the bridge between these two types of cyber attacks as they can either temporarily disable or physically destroy electronic devices, meeting the threshold for both types of attacks. Within this framework, cyberspace is related to the electromagnetic spectrum, as both are components of an integrated operational domain. Thus, when the PLA's *Science of Military Strategy* declares that "victory in war begins with victory in cyberspace" (战争的胜利先从网络空间开始), it entails the importance of EMP weapons, including HPMs (Science of Military Strategy, January 2020).

In an offensive context, HPM attacks could neutralize enemy communication networks, disrupt power grids, and disable electronic command systems, degrading an opponent's ability to coordinate a response. [5] For instance, in a Taiwan contingency, HPM strikes could target critical infrastructure such as command centers, radar installations, and missile defense systems, severely weakening Taiwan's ability to resist a PLA assault (Wuthnow et al., eds., 2023). PLA strategists emphasize EMP weapons as part of joint electronic and cyber warfare operations, whereby HPM attacks would be synchronized with cyberattacks to paralyze Taiwan's digital networks and power grids, forcing an early battlefield advantage.

From a defensive perspective, HPMs are an important anti-access weapon. Land-based HPMs can serve as a cost-effective air defense system against missiles and aircraft. Shipborne HPMs can deny enemy ships access to sea lanes. HPMs are also gaining importance as unmanned aerial vehicles (UAVs) and unmanned underwater vehicles (UUVs) play a larger role in modern combat, as HPMs can intercept swarms of small and digitally controlled attack systems.

Two primary goals lie behind the PRC's HPM research and development. The first is to achieve an asymmetric warfare advantage, using cost-effective means to neutralize high-tech adversaries without engaging in direct kinetic conflict. The second is to use paralyzing strikes (against radar air-defense systems, for example) to pave the way for conventional attacks ([Association of the United States Army](#), July 1, 2019). [6]

Russia Provides Object Lesson for Electronic Warfare

The PRC has closely studied recent conflicts to refine its HPM application theories. Since Russia's invasion of Ukraine beginning in February 2022, PLA research has emphasized the importance of early-stage cyber-electromagnetic dominance. [7] Russia's use of electronic warfare (EW) to disrupt Ukraine's communications systems and disable UAVs have shown Beijing how HPMs could shape the battlespace. [8] Although neither has used any HPM weapons specifically, lessons can still be gleaned from the overall EW doctrines of the parties to the current conflict.

Russia launched a wave of EW attacks against Ukrainian S-300 and SA-11 surface-to-air-missile batteries at the outset of the conflict, allowing for Russian missiles to strike Kyiv ([RUSI](#), November 7, 2022). Weapons like Russia's Khibiny pod, a jamming device attached to fighter jets, and Krasukha, a mobile land-based jamming device, were likely used in these EW strikes. The latter devices more recently have been effective at disrupting the fire and control systems of Ukraine's U.S.-supplied HIMAR missiles. Ukrainian anti-missile defenses were eventually reconstituted as Russia scaled back its EW attacks to avoid also impacting its own forces. Both sides have also used EW weapons for anti-access and denial purposes. Hand-held anti-drone guns that disrupt radio-wave signals between drones and the operators on the battlefield are common ([Hague Center of Strategic Studies](#), February 7, 2023). Russian drone and missile attacks on critical infrastructure such as power plants demonstrate the potential impact HPMs could have if used to disrupt such systems in a future conflict. These operations have left millions of Ukrainians experiencing regular blackouts and inconsistent energy supplies.

Taking lessons from the war in Ukraine, the PLA's emphasis on electromagnetic dominance has grown. Russia's use of EW weapons to disrupt Ukraine's C4ISR systems, particularly in the realm of aerial defense, has proven effective, as have defensive EW weapons degrading enemy UAV and guided missile attacks. PLA analysts also recognize the psychological impact of potential HPM attacks against critical infrastructure. EMP-induced blackouts and communication failures can create fear, confusion, and operational paralysis among adversaries. [9] Consequently, the PLA is integrating HPM weapons into broader cognitive and psychological warfare strategies to weaken enemy resolve.

Conclusion

PRC investment in HPM weapons reflects broader strategic objectives; namely, countering U.S. conventional superiority by asserting dominance in the cyber and electromagnetic space. The PRC's near monopoly of HPM patents, which ensures its dominance in research in this field, should be a particular concern for the United States. For those in the PLA's crosshairs, like Taiwan, PLA development of new aerial and ship-borne HPMs indicates greater offensive EMP capabilities that will threaten their C4ISR systems and critical infrastructure.

The PLA views EMP warfare, particularly HPMs, as critical to modern warfare. Further advances are crucial to Xi Jinping's vision of building a modernized PLA force by 2035. They also align with the PLA's informatized warfare doctrine, which emphasizes advancing cyber, space, and electromagnetic capabilities. So long as the PLA continues to believe the cyber-electromagnetic space is where the "victory of war begins," HPMs will play a critical role in any future conflict.

Notes

[1] Yang Ji, Li Yuchun, Liu Qiushi, et al. "Spatial Distribution Characteristics of High-altitude Nuclear Electromagnetic Pulse" (高空核電磁脈衝的空間分佈特徵). *Equipment Environmental Engineering* (裝備環境工程) 21, no. 10 (2024): 151–157.

[2] Li Bin, Xu Li, Wang Hao, Yin Junhui, Hu Quan, Hu Yulu, Yang Zhonghai. "Computational Electromagnetics and Electromagnetic Industry Software." *Journal of University of Electronic Science and Technology of China*, Vol. 53, No. 6, 2024, pp. 803–815. DOI: 10.12178/1001-0548.2024248.

[3] Pei, Gaofer, et al. "Simulated Analysis of Electromagnetic Pulse inside Ship Mast in High-power Electromagnetic Circumstance." *New Technology and New Processes* (新技術新工藝) no. 9 (2023): 29–37.

[4] Zhu, Yuanjiang, Yang Kang, Bao Qiuxiang, Xu Yu, Jiang Weixiang, and Song Shiqian. "Design and Verification of Broadband Multi-Beam Transmission Using Digital Arrays." *Journal of University of Electronic Science and Technology of China* 52, no. 6 (November 2023): 818–825. <https://doi.org/10.12178/1001-0548.2023077>.

[5] Liu, Lun'gang, et al. "Electromagnetic Pulse Weapons: Major Threats to Civil Defense Command and Communication Systems and Countermeasures." *National Defense Mobilization* (國防動員), no. 7 (2023): 8–9.

[6] Ge, Jun, et al. "Research on Intelligentized Warfare and Countermeasures for Combat Capability Construction." *National Defense Technology* (國防科技) 43, no. 6 (2022): 107–113.

[7] Li Xin Hui et al. "Analysis of the application of unmanned intelligent equipment in the Russian-Ukrainian conflict from the perspective of the kill chain" *Command Control & Simulation* (指揮控制與仿真) 46, no. 5 (2024): 6–12. For more on PLA lessons from Russia's war in Ukraine, see [China Brief](#), March 28.

[8] Zhang Yujie, Wang Wei, Xiong Zheng, and Dai Jingke. "Analysis of Joint Communications and Communication Confrontation in the Russia-Ukraine Conflict." *Defense Industry Conversion in China* (中國軍轉民) 1 (2024): 197–199.

[9] Yang Yang and Zhang Zhaoxin. "Electronic Warfare Systems in Future Wars and Their Impact." *Equipment* (裝備), no. 12 (2021): 44–45.



